

Article

Large-Scale Origin–Destination Prediction for Urban Rail Transit Network Based on Graph Convolutional Neural Network

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Abstract: Due to data sparsity, insufficient spatial relationships, and the complex spatial and temporal characteristics of passenger flow, it is very challenging to achieve a high prediction accuracy on Origin–Destination (OD) in a large urban rail transit network. This paper proposes a two-stage prediction network GCN-GRU, using a Graph Convolutional Network (GCN) with a Gated Recursive Unit (GRU). The GCN can obtain the adjacency relationship between different stations by selecting the adjacent neighborhoods and interacting neighborhoods of a station and capturing the spatial characteristics of the OD passenger flow. Then, an advanced weighted aggregator is employed to gather important information from the two above-mentioned types of neighborhoods to capture the spatial relationship of the network OD passenger flow and to perceive the sparsity and range of the OD data. On the other hand, the GRU can extract the temporal relationship, such as periodicity and other time-varying trends. The effectiveness of GCN-GRU is tested with a real-world urban rail transit dataset. The experimental results show that whether it is the OD passenger flow matrix of each period (one hour) on weekdays and weekends or the single-pair OD passenger flow between stations, the proposed GCN-GRU models perform better than the benchmark models. This study provides an important theoretical basis and practical applications for operators, thus promoting the sustainable development of urban rail transit systems.

Keywords: urban rail transit; large-scale OD passenger flow; prediction method; GCN-GRU



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1. Introduction

Urban rail transit, when available, has become instrumental in major cities because of its high capacity and efficiency in alleviating traffic congestion, good energy efficiency, safety, and environmental friendliness. Origin–Destination (OD) passenger flow refers to the spatial distribution of passenger traffic between the entrance and exit stations of the rail transit network during a certain period, which is usually expressed in the form of an OD matrix. As an important part of “smart urban rail transit”, large-scale OD passenger flow prediction provides support for intelligent transportation organizations. Urban rail transit OD passenger flow is the essential data for the rational formulation of transportation plans and provides the basis for decision-making of intelligent dispatching and emergency management. In-depth analysis of OD passenger flow’s spatial-temporal relationship, understanding its change patterns, and improving the accuracy and timeliness of its prediction can achieve more precise matching of the capacity and passenger flow demand and improve the efficiency and safety of the entire urban rail transit system. The Automatic Fare Collection (AFC) System of urban rail transit records the origin station, destination station, and time information of each trip of a passenger. By processing the record, OD passenger flow information can be obtained at any time granularity (such as 15 min, 30 min, or 1 h) in the network [1]. However, the AFC cannot obtain a passenger’s transfer station

recording in a rail transit network. As a result, OD passenger flow information cannot be obtained accurately for all passengers in real-time.

Whether it is road traffic flow or urban rail transit passenger flow, the OD prediction methods used in existing research can be roughly divided into three categories: linear, nonlinear, and combined. Linear prediction methods include the time series model [2,3], the Kalman filter model [4], etc. Non-linear prediction methods include nonparametric regression [5,6], neural network algorithm [7–10], and support vector machine [11–15]. In a study [13], an online support vector machine (OLSVM) model was constructed to predict the short-term in-station passenger flow in the metro. The research showed that the proposed model is suitable for abnormal passenger flow (such as on a holiday). Combined prediction methods include the wavelet decomposition combined model [16–19], Bayesian decomposition combination model [20–22], empirical mode decomposition combination model [23], neural network combination model [24,25], support vector machine combination model [26,27], etc. In another study [19], a combined model of SARIMA (Seasonal Auto-Regressive Integrated Moving Average Model) and SVM was established to predict the short-term passenger flow of a metro station. It was found that the proposed model can not only reflect the periodic change patterns but also realize real-time capture of the local characteristics of short-term passenger flows.

In recent years, prediction methods using deep neural networks have attracted increasing attention. Some studies [28–34] used Convolutional Neural Networks (CNNs) for traffic flow prediction on the road network and combined them with deep neural networks (e.g., Long Short-Term Memory (LSTM), GRU) to capture the temporal and spatial characteristics of traffic flows. Other studies [35,36] constructed deep networks based on stacked self-encoders in the deep learning framework to achieve traffic flow prediction. A graph convolutional neural network (GCN) was also used to solve the time series prediction problem in the traffic field [37–41]. Experiments show that the model can be trained faster and has fewer parameters, more effectively capturing the spatial-temporal correlation of traffic flows. A study [42] converted the urban railway transit network into a graph and used GCN to predict the in and out ridership of each station in the whole network. The results show that GCN can effectively capture the irregular spatial-temporal dependencies of the metro network.

Previous studies on passenger flow prediction of urban rail transit mainly focus on in and out passenger flow, and the research on OD passenger flow prediction is still in its infancy. Yao et al. [43] established a short-term OD matrix estimation state-space model to realize dynamic OD matrix estimation of urban rail transit under different time granularities based on the Kalman filter. Chen et al. [44] proposed a multi-model combination method for short-term OD estimation of urban rail transit with the diversion rate of passenger flows at different time granularities as a state variable. References [1,45–49] constructed a metro OD passenger flow prediction model based on LSTM, and the research results showed that LSTM can better capture temporal correlations in passenger flow. Zhang et al. [50,51] proposed the concept of OD attraction to solve the problem of OD data sparsity, and the research results showed that when the OD attraction level is “low”, a better prediction effect can be obtained. Hou et al. [52] proposed a neural network model based on GRU for short-term OD passenger flow estimation in urban rail transit, and the result demonstrated that the GRU model is simpler, converges faster, and has advantages in prediction accuracy and prediction stability compared with LSTM. Some studies [53–56] used GCN to predict OD flows in a large-scale urban transit network, and the results demonstrated the superiority of GCN by the data of metro passenger flow in different cities. Zheng et al. [57] and Zhang et al. [58] proposed the attention mechanism to enhance its prediction ability to capture both local and global temporal dependencies. Liu et al. [59] and Cheng et al. [60] realized real-time OD prediction through the online update algorithm.

In summary, there are still urgent problems to be solved in large-scale OD prediction of urban rail transit networks by scholars, which are mainly reflected in the following three aspects:

1. OD matrix dimension. With the acceleration of the urban rail transit network expansion in many cities, the OD passenger flow of the network becomes larger and larger, and the dimension of a single OD matrix can reach tens of thousands in a large network. For such a large OD matrix, the training of the prediction algorithm will be difficult, and the data scale will reach tens of millions or even hundreds of millions.
2. Spatial feature. Due to the complex spatial relationship between stations, large-scale OD prediction is still at the research stage, and the accuracy cannot meet the needs of practical applications. The spatial relationship between the stations in the network is another key issue that affects the prediction accuracy of the OD matrix. An in-depth study should be conducted on the adjacency of stations so that the prediction model can accumulate and select feature information from adjacent stations.
3. Data sparsity. The distribution of passenger flows between stations is imbalanced. As shown in Figure 1, the amount of OD between some stations is large, while that between some stations is very small. This study found that data sparsity has a great impact on the OD matrix prediction accuracy.

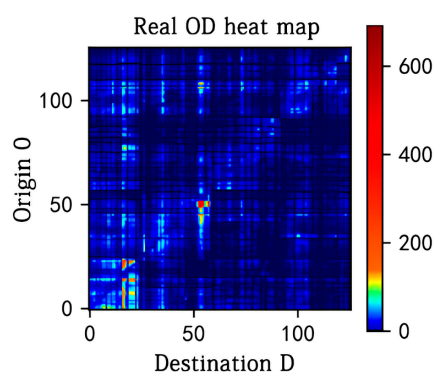


Figure 1. Heat map of OD passenger flow distribution of a metro network from 8:00 to 9:00 in the morning peak.

The urban rail transit large-scale OD passenger flow can be regarded as a high-order matrix. There are inherently complex connection relationships between the data, as well as much hidden information. Inspired by research in other related fields, the rail transit network can be converted into graph structure data, and the OD passenger flow relationship between different stations can be simulated by imitating the information transfer mode in the GCN.

In this study, the neighborhood of the station is selected, the spatial relationship between the stations is described, and the GCN is improved to accumulate and select important information from the neighborhood while sensing the sparsity and range of OD data. Then, by adopting the GRU in RNN to capture the temporal relationship of OD passenger flows, a large-scale OD passenger flow prediction system for urban rail transit with GCN-GRU as the core is constructed, which provides a decision basis for passenger flow congestion prediction and emergency coordination and coordinated control. Also, it is an application of passenger traffic flow theory and a useful supplement to artificial intelligence technology applications in transportation engineering.

The main contributions of this research are summarized as follows:

1. Existing recurrent neural networks (such as LSTM and GRU) can well capture the temporal relationship of passenger flows or traffic flow data, but the extraction of spatial relationships is limited to nearby areas with close geographical locations. With the construction of satellite cities and the separation between jobs and residences, there is a strong interaction of passenger flow between distant stations. Therefore, this paper uses the indicator of passenger flow to construct the interactive neighborhood of the station and uses the distance between the stations to construct the adjacent

neighborhood of the station to comprehensively describe the spatial relationships between stations.

2. GCN is employed to solve large-scale OD passenger flow prediction problems. Large-scale OD passenger flow prediction in urban rail transit is challenging. Considering the advantages of GCN in processing graph structure data in other fields in recent years, this paper converts the urban rail transit network into a graph and combines GRU units to demonstrate that the GCN-GRU model can obtain satisfactory accuracy and feasible efficiency for large-scale OD passenger flow prediction through experiments.
3. An advanced weighted aggregator is proposed to sense the sparsity and range of OD data. For high-order OD passenger flow matrices, the problem of data sparsity has always been an important factor affecting the prediction effect. In this paper, by improving the naive aggregator in GCN and taking the distance and passenger flow between the stations and the neighborhoods as the weight factors, two advanced weighted aggregators are constructed in parallel, which can selectively focus on the important neighborhood of the station to perceive the sparsity and scope of the data.

The remaining sections are organized as follows. Section 2 introduces the related theory and explains the relevant definitions. The model framework and the evaluation indicators are described in detail in Section 3. The experimental details and results are presented in Section 4, and Section 5 concludes this paper.

2. Theoretical Basis

2.1. GCN Theory

In recent years, to process a large amount of graph structure data, and driven by the development of deep neural networks, Graph Neural Network (GNN) has emerged. As an important variant of GNN, GCN has been used in many fields, such as protein sequence, natural language processing, etc., and it has achieved good results. Meanwhile, related research shows that GCN has a powerful application value in the field of traffic flow prediction, which can fully capture the complex spatial dependencies of traffic flows and significantly improve prediction accuracy. The above fields have similar graph structure data to the rail transit network, which provides theoretical support for the research on large-scale OD passenger flow prediction.

Define a graph $G = (V, E, A)$, where V is the node set of the graph, E is the edge set of the graph, and $A \in R^{n \times n}$ represents the adjacency matrix of the network. Then, the GCN convolution operation is defined in Formulas (1) and (2).

$$H^{l+1} = f(H^l, A) = \sigma(\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^l W^l + b^l) \quad (1)$$

$$H^{l'} = \hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^l \quad (2)$$

where $\hat{A} = A + I$, $A \in R^{n \times n}$ is the adjacency matrix, $I \in R^{n \times n}$ is the identity matrix, $\hat{D}$ is the diagonal node degree matrix of $\hat{A}$, W^l is the parameter matrix of the l -th layer, and b^l is the bias vector of the l -th layer; $H^l \in R^{n \times t}$ is the feature matrix of the l -th layer, where n is the number of nodes, t is the number of features per node; $H^{l'} \in R^{n \times t}$ is the feature matrix of the l -th layer containing topological information; $\sigma(\cdot)$ is the activation function.

2.2. GRU Theory

Cho K, Merriënboer B V, and Gulcehre C [61] proposed Gated Recurrent Unit (GRU) in 2014, which is an improvement from LSTM. The difference between GRU and LSTM is that GRU merges the forget gate and the input gate into the update gate, and it also merges the memory unit and the hidden layer into a reset gate. Extensive experiments have demonstrated that GRU is simpler in structure and has fewer parameters than LSTM. Moreover, GRU has no obvious gap in performance compared with LSTM in practical applications, and GRU may even perform better on some tasks.

The GRU model is shown in Figure 2, where the yellow rectangle represents the learned neural network layer, the pink circle represents the operation, and the arrow represents the vector transmission process.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (3)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (4)$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t]) \quad (5)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (6)$$

where $\cdot$ represents matrix multiplication; $*$ is the Hadamard product; $[]$ means that the two matrices are connected; σ is the sigmoid function; x_t is the input value at time t ; h_{t-1} is the output hidden layer state at time $t - 1$; $\tilde{h}_t$ is the candidate set; h_t is the output state at the time t ; z_t is the update gate to control the number of states of the hidden layer at the previous time is updated to the current hidden layer; r_t is the reset gate to control the number of states of the hidden layer at the previous time is updated to the current candidate hidden layer; W is the weight matrix.

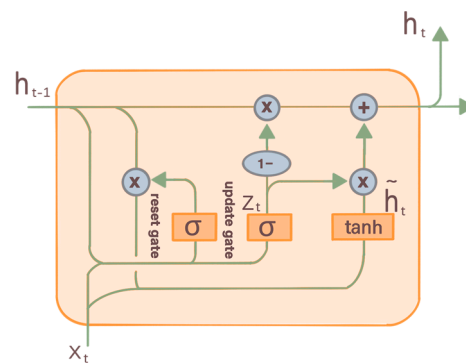


Figure 2. GRU network structure.

2.3. Definitions

Definition 1 Time Lag. This paper divides the time average into consecutive t lags, which is represented as $\{LAG_1, LAG_2, \dots, LAG_k, \dots, LAG_t\}$, and the interval between any two consecutive lags is constant.

Definition 2 OD Passenger Flow. At a certain period, the starting station (Origin) is denoted as i , the destination station (Destination) is denoted as j , and $i, j = 1, 2 \dots N$ (N is the number of stations in the network), and the amount of passenger traffic travel between the in station and out station is denoted as A_{i-j} . Then, this paper defines the OD passenger flow as an N -order asymmetric matrix $A \in M^{N \times N}$, and $A_{i-j} \in A$.

Definition 3 OD Passenger Flow Forecast. For t time lags, given a series of t real values of OD passenger flow matrices $\{A_1, A_2, \dots, A_k, \dots, A_t\}$ and some available potential influencing factors, such as weather information, etc., OD passenger flow prediction is to calculate the OD passenger flow matrix predictions $\hat{A}_{t+1}$ at the next time lag LAG_{t+1} .

3. Model Framework

Effective feature extraction is the key to obtaining a better prediction effect. More importantly, to describe the spatial dependencies between stations, it is crucial to select neighborhoods that are closely related to the current station. After obtaining the OD matrix, this paper extracts features and determines the feature vectors. Then, the adjacency and interaction characteristics between stations are considered, and the rules are defined for selecting neighborhoods to obtain a neighborhood set, which lays a theoretical foundation for the model construction of OD passenger flow prediction.

The data sparsity of the OD matrix is a key problem affecting the prediction accuracy, which cannot be solved by the naive aggregator in GCN. Meanwhile, the OD passenger flow is time-series data, and it is equally important to capture temporal dependencies. Therefore, based on the above feature extraction and station neighborhood selection, this paper improves the weighted aggregator in GCN and proposes an advanced weighted aggregator. The proposed weighted aggregator can sense the sparsity and range of the data, accumulate and select feature information from important neighborhoods of stations, and generate new embedding vectors. Based on this, the gated recurrent unit (GRU) in the RNN is introduced to capture the temporal correlation to accurately predict the OD matrix.

Figure 3 shows the general process of the GCN-GRU model framework. First, the OD matrix can be extracted from the passenger transaction information (i.e., in and out stations recorded by the AFC system). Then, feature extraction is performed on the above data to obtain feature vectors. Subsequently, geographical adjacency neighborhoods and passenger flow interaction neighborhoods between stations in the network are selected according to neighborhood selection rules. Finally, the neighborhoods are inputted into the GCN network to calculate the embedded vector and input into the GRU unit to obtain the predicted OD matrix.

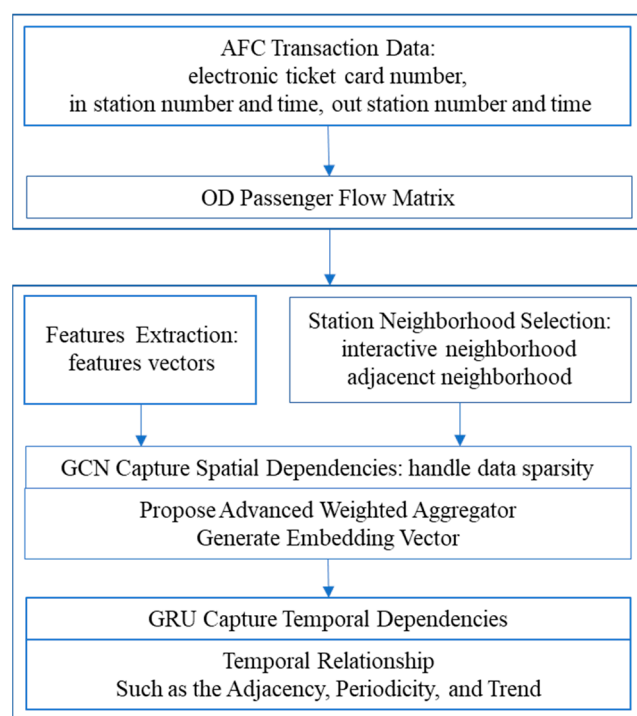


Figure 3. Large-scale OD passenger flow prediction process based on GCN-GRU.

3.1. GCN Network Framework

In this paper, a GCN-GRU model is constructed to predict the large-scale OD passenger flow matrix of urban rail transit. The structure of the GCN framework is shown in Figure 4. Assuming that an urban rail transit network contains N stations. Two input features are determined, where the first one is the interaction feature, and the second one is the adjacency feature: (1) according to Section 3.1.1, the interaction neighborhood of each station is determined and represented as the OD matrix $A_1, A_2, \dots, A_k, \dots, A_t$ at multiple consecutive time lags $1 \sim t$. Then, according to the advanced weighted aggregator Formula (12) in Section 3.1.3, the interaction weight is calculated to form a new interactive feature vector. The feature vector is input into the GCN network together with the OD matrix to obtain an interactive embedding vector of size $(B \times t \times N \times E)$, where B represents the batch size, and E represents the embedding dimension. (2) According to Section 3.1.2, the adjacency neighborhood of each station is determined, and the shortest line distance matrix,

i.e., the first-order adjacency matrix adj_{i-j} considering the distance between stations, is used. Then, according to the advanced weighted aggregator Formula (13) in Section 3.1.3, the adjacent weight is obtained to form a new adjacent feature vector. The feature vector is inputted into the GCN network together with the OD matrix to obtain an adjacent embedding vector of size $(B \times t \times N \times E)$. Afterward, the above two input features are used to capture the topological relationship of the spatial-temporal interaction relationship between stations, extract the high-order spatial-temporal features hidden in the data, and obtain two feature matrices of size $(B \times t \times N \times E)$. Finally, the above two input features are integrated and flattened into a matrix of size $(B \times t \times N \times 2E)$ and sent to the GRU network.

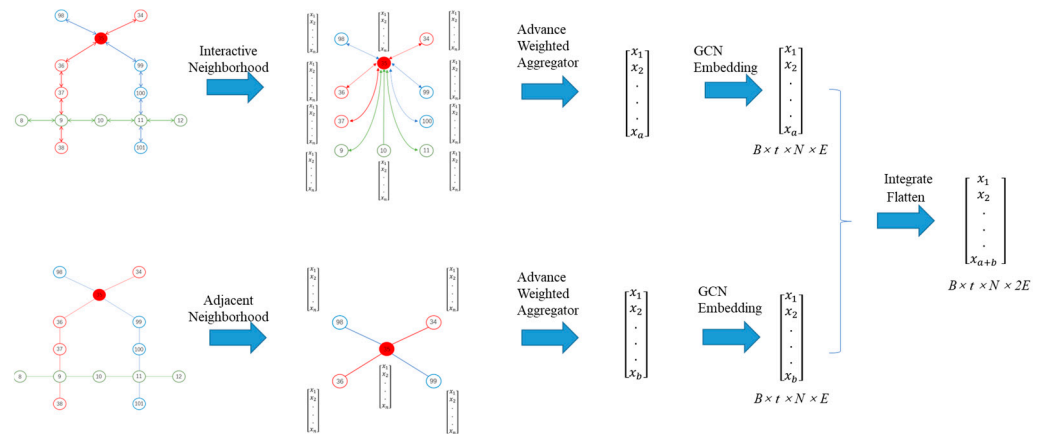


Figure 4. GCN network framework diagram.

Due to the limitation of GCN on small-scale OD passenger flow matrices, this paper proposes two neighborhood functions to model the spatial relationship of stations, namely, interactive neighborhood and adjacent neighborhood. The two neighborhoods are used to capture the passenger flow interaction strength between the origin and destination stations in the rail transit network and to measure the intrinsic closeness between a station and its neighbors. Figure 4 illustrates the aggregation process of these two neighborhoods.

3.1.1. Interactive Neighborhood

A rail transit network contains N stations. If there is at least one OD passenger flow between two stations, one can be defined as an interactive neighborhood of the other, which is used to model the passenger flow intensity between stations.

At any time lag $k = 1, 2, \dots, t$, the interaction neighborhood of the station can be obtained by Formula (7). It can be seen from Formula (7) that since the OD passenger flow is time-sensitive, the number of interaction neighborhoods of different stations in different time lags is not constant. Therefore, it is crucial to consider the interaction relationship of different stations in different time lags.

$$\alpha_i^k = \{j = 1, 2, \dots, N \mid A_{i-j} > 0 \parallel A_{j-i} > 0, A_{i-j} \in A_k, A_{j-i} \in A_k\} \quad (7)$$

3.1.2. Adjacent Neighborhood

The adjacent neighborhood between stations mainly refers to geographical closeness. That is, there is a direct line connection between the stations, which is used to measure the intrinsic closeness between the current station and its adjacent stations.

The adjacent neighborhood of the station i can be obtained by Formula (8). It can be seen from Equation (8) that the 1st-order adjacency relationship between two stations is 1, which can be defined as an adjacent neighborhood.

$$\beta_i = \{j = 1, 2, \dots, N \mid adj_{i-j} = 1 \parallel adj_{j-i} = 1, A_{i-j} \in adj, A_{j-i} \in adj\} \quad (8)$$

The 1st-order adjacency refers to the direct line connection between the study station and its adjacent stations. Therefore, based on the topology structure of the rail transit network, the 1st-order adjacency matrix between stations can be constructed, and it is a symmetric matrix. The 1st-order adjacency matrix adj can be represented as follows:

$$adj = \begin{bmatrix} adj_{1-1} & adj_{1-2} & \dots & adj_{1-N} \\ adj_{2-1} & adj_{2-2} & \dots & adj_{2-N} \\ \vdots & \vdots & \ddots & \vdots \\ adj_{N-1} & adj_{N-2} & \dots & adj_{N-N} \end{bmatrix} \quad (9)$$

where, the first-order adjacency relationship of any two stations i and j can be represented by the following formula:

$$adj_{i-j} = \begin{cases} 1, & \text{When station } i \text{ and station } j \text{ have a line connection} \\ 0, & \text{When station } i \text{ and station } j \text{ have no line connection} \end{cases} \quad (10)$$

3.1.3. Advanced Weighted Aggregator

In this paper we train an aggregator function that learns to accumulate and select feature information from the neighborhoods of the stations by aggregating interactive neighborhood α and adjacent neighborhood β to deduce the vector representation for each station, instead of training a different embedding vector for each station. Before introducing the advanced weighted aggregator proposed in this paper in detail, the adopted naive aggregator form is first briefly introduced [62]:

$$v_i = \sigma \left(W \cdot \text{MEAN} \left(\{v_i'\} \cap \{v_j', j \in N_i\} \right) \right) \quad (11)$$

where v_i is the embedding vector of station i , W is the weight matrix, σ is the sigmoid function, $\text{MEAN}(\cdot)$ represents the mean value, and v_j is the embedding vector of station j . In the naive aggregation method, v_i is obtained by computing the mean value of the feature vectors v_j' and their neighbors $j \in N_i$ and concatenating it to the previous features v_i' . However, despite the fact that some variants of the basic aggregator have been proposed, such as pooling aggregator and LSTM aggregator, the existing graph convolutional aggregation methods cannot fully capture the relationship between different stations in the OD passenger flow matrix scenario and cannot construct the OD passenger flow spatial relationship model because these aggregators cannot distinguish their importance when fusing all the features of each station's neighborhood. Intuitively, in an interacting neighborhood set, the degree of interaction strength of neighborhood stations should affect the aggregation process because it retains representative travel patterns. Meanwhile, the closer the geographic distance between the two stations, the more similar their properties will be. Based on this, this paper proposes an advanced weighted aggregator that selectively focuses on important neighborhoods for station embedding.

Denote $flow(i \leftrightarrow j)$ (from i to j or from j to i) as the weighting factor of the aggregator, i.e., the passenger flow between station i and interactive neighborhood j ($j \in \alpha_i^k$). Therefore, the advance weighted aggregator of the interaction neighborhood is expressed as follows:

$$p_i^k = \sigma \left(W_p \cdot \left(f_i^k + \sum_{j \in \alpha_i^k} \frac{flow(i \leftrightarrow j)}{\sum_{j \in \alpha_i^k} flow(i \leftrightarrow j) + \epsilon} f_j^k \right) \right) \quad (12)$$

where ϵ is a very small value close to zero in the case $flow(i \leftrightarrow j) = 0$; p_i^k represents the embedding vector of the interaction neighborhood of station i at time k ; W_p is the trainable weight matrix; f_i^k and f_j^k respectively represent the feature vectors between station i and neighborhood j ($j \in \alpha_i^k$) before the interaction weighting.

Denote $\frac{1}{dis_{i-j}}$ as the weighting factor of the aggregator, i.e., the inverse of the distance between the station i and its neighbors j ($j \in \beta_i$). Therefore, the advance weighted aggregator of adjacent neighborhoods can be denoted as follows:

$$q_i^k = \sigma \left(W_q \cdot \left(f_i^k + \sum_{j \in \beta_i} \frac{(\frac{1}{dis_{i-j}})}{\sum_{j \in \beta_i} (\frac{1}{dis_{i-j}})} f_j^k \right) \right) \quad (13)$$

where q_i^k represents the embedding vector of the adjacent neighborhood of station i at time k ; W_q is the trainable weight matrix; f_i^k and f_j^k , respectively, represent the feature vectors between station i and neighborhood j ($j \in \beta_i$) before the adjacent weighting.

It is worth noting that the advanced weighted aggregators p_i^k and q_i^k are both time-varying because the feature vectors of each station and its interactive neighborhood are time-varying.

The two embedding vectors p_i^k and q_i^k , respectively, represent the interactive neighborhood and adjacent neighborhood feature information; the embedding vector r_i^k represents the weather and time feature information of the station. Finally, the above three embedding vectors are connected to represent v_i^k as the final embedding of station i at time k :

$$v_i^k = [p_i^k, q_i^k, r_i^k] \quad (14)$$

3.2. GRU Network Framework

The matrices are integrated into a matrix of size $(B \times T \times N \times 2E)$ and then input into the GRU recurrent neural network in the order of $1 \sim t$; finally, the entire predicted feature vector of size $T + 1$ is output, and a series of decoding operations and convolution operations are performed to obtain the final prediction results. Compared with the real OD passenger flow matrix at time $t + 1$, the reverse gradient values are calculated only for the non-zero values in the OD passenger flow matrix by the Masked MSE Loss function. When the number of epochs reaches the upper limit, the update process stops immediately, as shown in Figure 5. Through a superimposed network model, such as GCN-GRU, the spatial-temporal correlation of data is fully considered to improve the prediction accuracy.

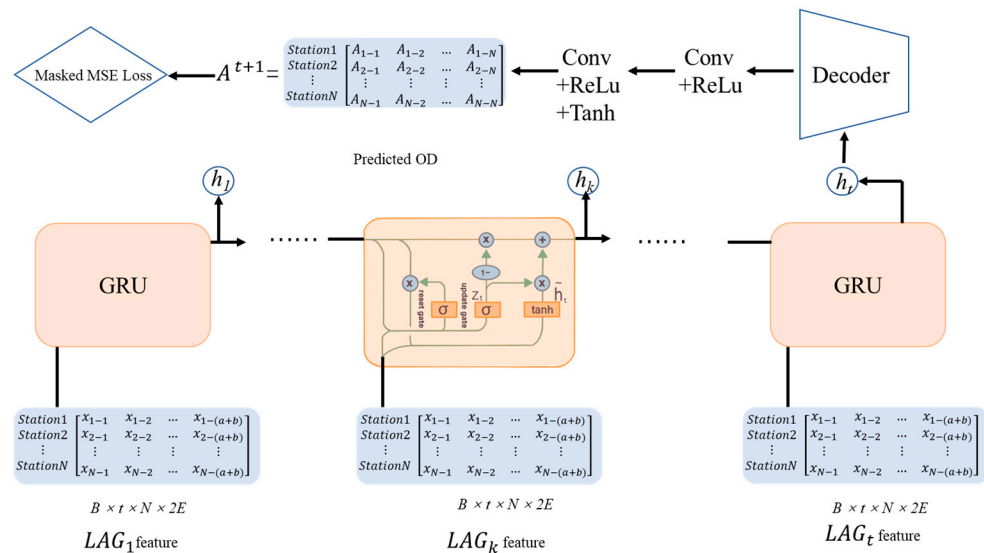


Figure 5. GRU network framework diagram.

3.2.1. Loss Function

This paper uses a masked MSE loss function (*M-Loss*) [50] shown in Formula (15).

$$M - Loss = \frac{1}{(n \times n)_{no-mask}} \sum_{i,j} mask \times \left(A_{i-j}^t - \hat{A}_{i-j}^t \right)^2 \quad (15)$$

where *M-Loss* is the masked mean-squared error; $(n \times n)_{no-mask}$ indicates the OD numbers that are not masked; A_{i-j}^t is the actual value of the passenger traffic traveling from station *i* to station *j* at time LAG_t ; $\hat{A}_{i-j}^t$ is the corresponding predicted value, and the mask is a matrix file with values of zeros and ones, which indicates whether the corresponding OD flows are masked or not. Note that, if the values are masked, the errors are not backpropagated, which can significantly improve the prediction performance of OD flows that are not masked. The mask highlights the OD pairs with higher passenger flows. Only the errors of important flows (i.e., OD pairs with higher passenger flows) will be backpropagated here.

3.2.2. Optimization Strategy

To optimize the parameters in the GCN-GRU model, this paper uses the mean square error (MSE) defined in 3.3 as the criterion to minimize and uses stochastic gradient descent (SGD) to update the parameters. Specifically, Adam [63], a variant of SGD, is adopted to optimize the parameters in the predictive model.

3.3. Evaluation Indicators

To compare the prediction effects of different models, four performance indicators commonly used in prediction algorithms are taken to evaluate the model performance, namely mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE), and Mean Absolute Percent Error (MAPE). The prediction effect of the model is tested on the test set, and the predicted value is compared with the actual value. The smaller the value of the four indicators, the smaller the error of the model and the higher the prediction accuracy of the model.

1. MSE

$$MSE = \frac{1}{N * N * S} \sum_{k=1}^S \sum_{i=1}^N \sum_{j=1}^N \left(A_{i-j}^{t+k} - \hat{A}_{i-j}^{t+k} \right)^2 \quad (16)$$

2. RMSE

$$RMSE = \sqrt{\frac{1}{N * N * S} \sum_{k=1}^S \sum_{i=1}^N \sum_{j=1}^N \left(A_{i-j}^{t+k} - \hat{A}_{i-j}^{t+k} \right)^2} \quad (17)$$

3. MAE

$$MAE = \frac{1}{N * N * S} \sum_{k=1}^S \sum_{i=1}^N \sum_{j=1}^N \left| A_{i-j}^{t+k} - \hat{A}_{i-j}^{t+k} \right| \quad (18)$$

4. MAPE

$$MAPE = \frac{1}{N * N * S} \sum_{k=1}^S \sum_{i=1}^N \sum_{j=1}^N \left| \frac{A_{i-j}^{t+k} - \hat{A}_{i-j}^{t+k}}{A_{i-j}^{t+k}} \right| \quad (19)$$

where *S* is the number of predicted time intervals, *N* is the number of stations in the rail transit network, A_{i-j}^{t+k} is the actual value of the passenger traffic traveling from station *i*

to station j at the time LAG_{t+k} , and $\hat{A}_{i-j}^{t+k}$ is the predicted value of the passenger traffic traveling from station i to station j at the time LAG_{t+k} .

4. Experiments

4.1. Data Collection

In data collection, the first task is to select a certain urban rail transit network as the research object, including the station number and the distance between stations to form the network topology. As shown in Figure 6, the experimental network includes four lines and 126 stations. The average distance between the stations is about 1200 m.

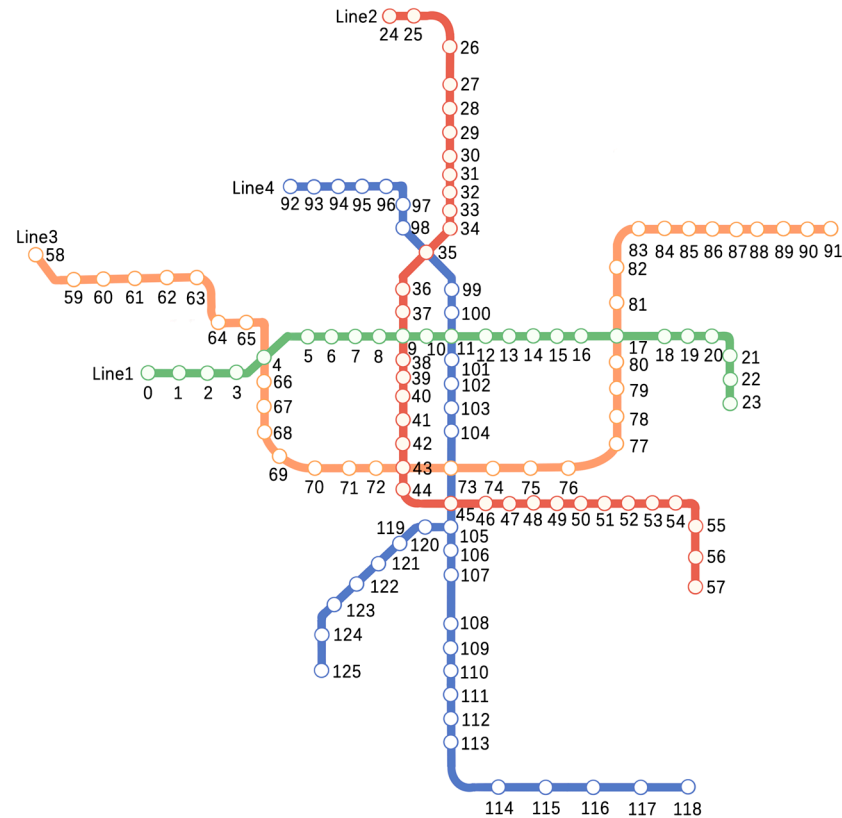


Figure 6. Urban railway transit network topology.

Then, the passenger in and out station transaction data recorded by the city's AFC system from 1 March to 25 April 2021 (8 weeks in total) are taken, including electronic ticket card number, in station number, in station time, out station number, out station time, etc. This paper takes 60 min as the statistical time interval and counts data to obtain the OD passenger flow matrix. There are 24 OD matrix observations of size 126×126 per day, forming a total of 1344 consecutive OD passenger flow matrix sequences of size 126×126 .

The dimension of an OD passenger flow matrix in this city can reach 10,000 or even larger. Meanwhile, there is an inherently complex connection between the data and there is a lot of hidden information. To use GCN to realize large-scale OD passenger flow prediction, the above data are processed, and the urban rail transit network is converted into a graph structure, where the nodes represent the stations, the edges represent lines, and edge weights represent OD passenger flows. The OD passenger flow relationship between different stations is simulated by imitating the information transfer mode in GCN.

4.2. GCN-GRU Model Construction and Performance Evaluation

4.2.1. Model Configuration

To facilitate prediction, the obtained OD passenger flow data set is divided into a training set, a validation set, and a test set at the ratio of 7:2:1. After the division, there are

925 OD tables in the training set, 132 OD tables in the validation set, and 264 OD tables in the test set. Then, a time window function is constructed to slide and intercept the data. One time step represents 1 h, and 24 time steps are used as input to predict the OD passenger flow data of one time step in the future. The dimension of the input data is $126 \times 126 \times 24$, i.e., the OD matrix (126×126) interacts at 24-time steps of 126 stations; the dimension of the output data is $126 \times 126 \times 1$, i.e., the OD matrix (126×126) interacts at the 1-time step of 126 stations in the future.

After parameter adjustment, this paper uses Adam as the optimizer. The initial learning rate is set to be 0.001, the weight decay rate 1×10^{-6} , the batch size 16, and the gradient clipping technology is used to prevent gradient explosion. Additionally, the early stopping technique controls whether the training of the model should be terminated, and the maximum number of epochs is set to 100.

4.2.2. Benchmark Model

To verify the effectiveness and rationality of the GCN-GRU model in solving the large-scale OD passenger flow prediction problem of urban rail transit, this paper compares the GCN-GRU model with the following benchmark models:

1. Historical average model (HA). In this paper, the HA model is used to predict the OD passenger flow matrix, that is, the historical average of passenger traffic flow between the origin and destination stations is used as the predicted value.
2. Convolutional long short-term memory network model (ConvLSTM) [64]. The ConvLSTM model adopts a general structure. Its input dimension is 1, the hidden dimension is 8, the kernel size is 3×3 , and the number of layers is 1. Meanwhile, the hyper-parameter settings (learning rate, batch size, etc.) are consistent with those of the GCN-GRU model in this paper.

4.2.3. Predictive Performance Evaluation

Table 1 summarizes the prediction performance of all the prediction models for the test set 264 OD passenger flow matrices of size 126×126 , i.e., the prediction matrix $\{\hat{A}_{t+1}, \hat{A}_{t+2}, \dots, \hat{A}_{t+264}\}$ and the actual matrix $\{A_{t+1}, A_{t+2}, \dots, A_{t+264}\}$. Then, the four performance indicators are calculated by Formulas (16) to (19). Due to space limitations, Tables 2–5 show the performance measures of all prediction models for 24 OD passenger flow matrices of size 126×126 on certain prediction days. Figure 7 shows the reduction of the prediction errors of the GCN-GRU model on each prediction day compared with the HA model; the reduction of MSE is more than 50%, the reduction of RMSE is 35% on average, the reduction of MAE is about 20%, and the reduction of MAPE is about 15%, all very significant percentage-wise. Especially on 17 April and 24 April 2021, the GCN-GRU model proposed in this paper achieves the best prediction effect, indicating that the proposed model can be applied to OD passenger flow prediction on different characteristic dates such as workdays and weekends.

Table 1. Comparison of prediction performance evaluation indicators of various models.

Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	10.645	3.263	0.942	0.551
ConvLSTM	33.172	5.76	2.16	1.011
GCN-GRU	5.893	2.428	0.764	0.485

Table 2. Comparison of prediction performance evaluation indicators of various models on the weekday (15 April 2021).

Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	4.899	2.213	0.816	0.493
ConvLSTM	32.514	5.702	2.165	1.016
GCN-GRU	2.337	1.529	0.645	0.434

Table 3. Comparison of prediction performance evaluation indicators of various models on the weekday (16 April 2021).

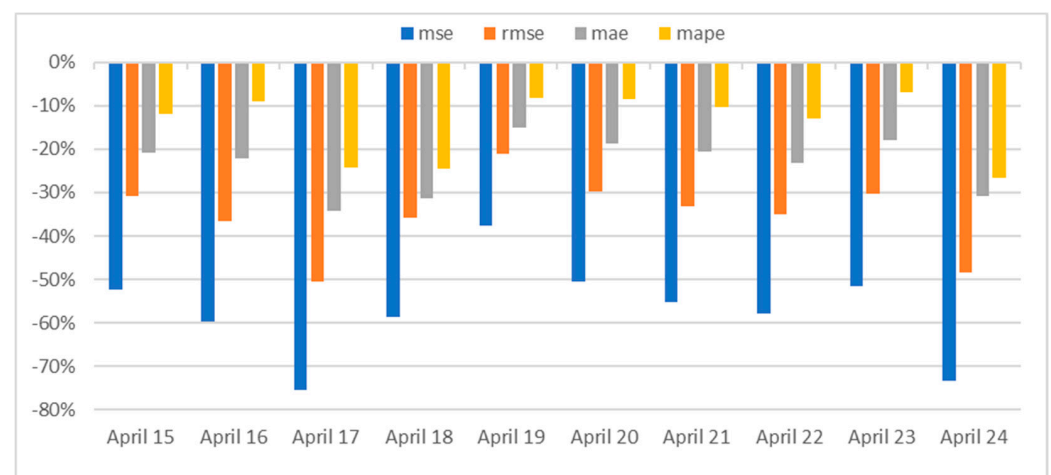
Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	7.333	2.708	0.906	0.48
ConvLSTM	37.703	6.14	2.274	0.984
GCN-GRU	2.945	1.716	0.706	0.437

Table 4. Comparison of prediction performance evaluation indicators of various models on the weekend (24 April 2021).

Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	14.428	3.798	1.088	0.698
ConvLSTM	18.01	4.244	1.963	0.991
GCN-GRU	3.855	1.963	0.754	0.513

Table 5. Comparison of prediction error indicators of each model from station 100 to station 35.

Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	7064.073	84.048	39.977	0.323
ConvLSTM	10638.81	103.145	52.958	0.46
GCN-GRU	2124.826	46.096	26.236	0.253

**Figure 7.** The reduction percentage of prediction error indicators of GCN-GRU and HA on different days.

Based on results from Tables 1–5, the following observation can be obtained:

The ConvLSTM model obtains the worst predictive performance among all models, and it performs significantly worse than HA. LSTM achieves better performance in capturing the temporal features of the passenger flow, and it ignores the spatial features. Although ConvLSTM uses convolutional layers to model spatial attributes, it is not suitable for OD matrix prediction, and this is probably because the spatial relationships between adjacent elements in the OD passenger flow matrix have no meaning.

The HA model does not achieve the optimal prediction performance, but it performs stably. This prediction result indicates that if the properties of a problem cannot be captured, a simple model might be a reasonable choice.

Whether it is a total of 264 time periods in the next 11 days or a total of 24 time periods on a workday or weekend, the GCN-GRU network achieves the best comprehensive results among all models in terms of the error indicators. The results verify that it is more reasonable to convert the large-scale OD passenger flow matrix modeling of urban rail transit into a graph structure.

Tables 5–7 show the prediction performance of all prediction models for the passenger traffic flow between some origin and destination stations, i.e., the predicted passenger flow value $\hat{A}_{i-j}^{t+1}, \hat{A}_{i-j}^{t+2}, \dots, \hat{A}_{i-j}^{t+264}$ and the actual passenger flow value $A_{i-j}^{t+1}, A_{i-j}^{t+2}, \dots, A_{i-j}^{t+264}$. Then, the four prediction error indicators are obtained according to Formula (16) to Formula (19). From the 10 groups of OD passenger flow with the largest historical average passenger flow value, Figure 8 shows the reduction of the prediction error indicators of the proposed GCN-GRU model and the HA model. It can be seen that the prediction performance of the GCN-GRU network has different degrees of improvement in most cases, especially from station 100 (commercial complexes) to station 35 (comprehensive transportation hubs), where the two error indicators of MSE and RMSE decrease by 70% and 45%, respectively. From station 51 (residential agglomerations) to station 53 (office agglomerations), the two error indicators of MAE and MAPE decreased by 59% and 46%, respectively. This also verifies that the model proposed in this paper is suitable for OD passenger flow prediction between multiple types of stations, such as comprehensive transportation hubs and commercial complexes, residential agglomeration areas, and office agglomeration areas.

Table 6. Comparison table of prediction error indicators of each model from station 51 to station 53.

Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	3501.941	59.177	17.718	0.609
ConvLSTM	11356.04	106.565	39.262	1.531
GCN-GRU	1237.103	35.172	7.307	0.327

Table 7. Comparison table of prediction error indicators of each model from station 23 to station 16.

Model	MSE/(Person/h) ²	RMSE/Person/h	MAE/Person/h	MAPE/%
HA	1487.038	38.562	14.643	0.443
ConvLSTM	7909.529	88.936	32.767	0.642
GCN-GRU	498.548	22.328	6.112	0.233

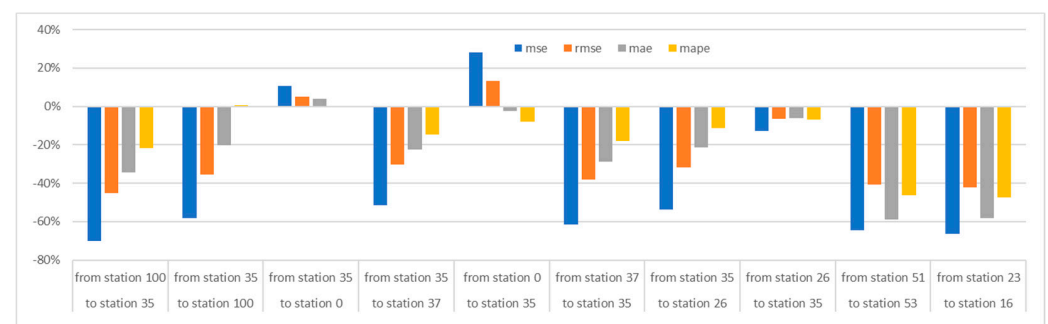


Figure 8. The reduction of prediction error indicators of GCN-GRU and HA on different stations.

4.2.4. OD Passenger Flow Prediction Result Visualization

To more intuitively show the prediction effect of the GCN-GRU model, this paper takes the predicted heat map and the real heat map of the city's rail transit network for comparison during the morning peak hours, evening peak hours on weekdays, and the

periods on weekends. The comparison results are shown in Figures 9–11, where the dark blue represents that the passenger traffic flow between the stations is close to 0, and the gradual change of the color bar from green to yellow and dark red represents that the number of passenger traffic flows between the stations has gradually increased. It can be seen from Figures 9–11 that the heat map of the GCN-GRU model is close to the actual heat map, most of the OD ridership values are small, and very few OD ridership are large.

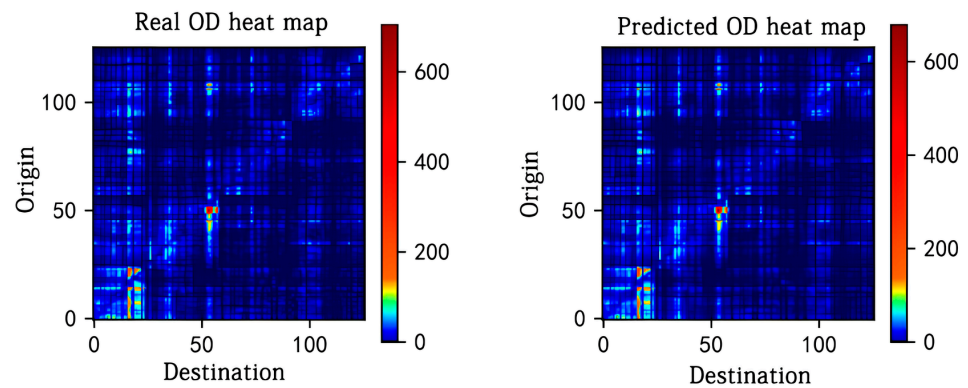


Figure 9. The prediction effect picture of OD passenger flow GCN-GRU from 8:00 to 9:00 in the morning peak of a weekday (April 20).

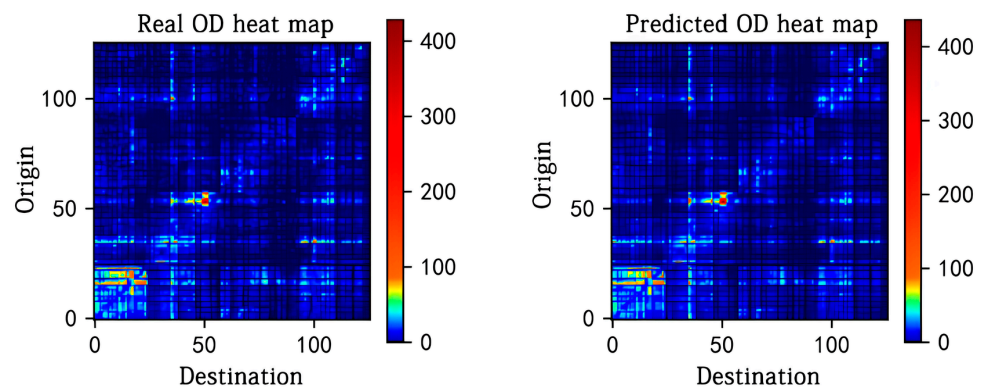


Figure 10. The prediction effect picture of OD passenger flow GCN-GRU from 18:00 to 19:00 in the evening peak of a weekday (April 23).

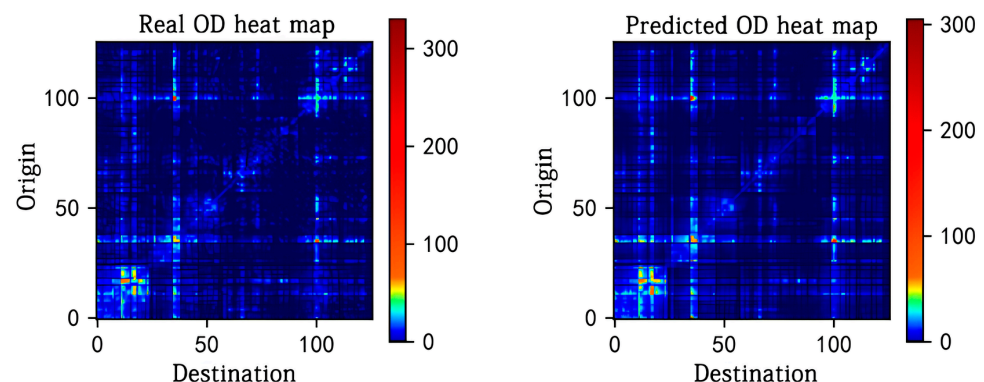


Figure 11. The prediction effect picture of OD passenger flow GCN-GRU from 14:00 to 15:00 on a weekend (April 24).

From Figure 9 to Figure 11, the following observation can be obtained:

Figure 9 shows that, during the morning peak, people tend to travel from residential agglomerations (such as stations 7, 14, 23, 49, 50, and 51) to office agglomerations (such as stations 16, 21, 53, 54, and 57), which are represented as the red block in the figure. The

commuter passenger flow occupies the dominant position, and the largest OD passenger flow in the morning peak hours from station 51 to station 53 reaches 687, as shown in Table 8.

Table 8. Details of the 10 largest groups of OD passenger flows in the morning peak on weekdays (April 20).

Serial Number	Hourly OD Passenger Flow (Person/h)	Starting Station Number	Arrival Station Number
1	687	51	53
2	684	50	53
3	559	51	54
4	536	50	54
5	516	49	53
6	494	23	16
7	426	14	16
8	338	51	57
9	325	7	16
10	308	14	21

Figure 10 and Table 9 show that during the evening peak, people tend to return from office agglomeration areas to residential agglomeration areas. The passenger flow traffic directions of the origin and destination stations with large passenger flows are almost opposite at the morning peak, and the return commuter passenger flow occupies a dominant position. However, compared with the morning peak, the overall passenger flow concentration is more scattered at the evening peak, and the largest OD passenger flow is 430 persons from station 53 to station 51. In addition, commercial complexes (such as stations 17 and 100) also attract a lot of ridership for leisure and entertainment at night. Moreover, stations 23 and Stations 14 are both residential agglomerations, but Station 23 is the terminal station of line 1, and nearby companies arrange special shuttle buses to send their employees to Station 23 to transfer to rail transit to go home, thus forming passenger flow between station 23 and station 14 in the evening peak.

Table 9. Details of the 10 largest groups of OD passenger flows in the evening peak on the weekday (April 23).

Serial Number	Hourly OD Passenger Flow (Person/h)	Starting Station Number	Arrival Station Number
1	430	53	51
2	377	53	50
3	350	54	51
4	318	54	50
5	288	53	49
6	254	35	100
7	239	16	23
8	211	23	14
9	195	23	17
10	190	57	51

It can be seen from Figure 11 and Table 10 that the passenger flow on weekends is not as concentrated as that in the morning and evening peaks, the red blocks are significantly reduced, the largest OD passenger flow is 282 persons, and the larger OD passenger flow is concentrated in the old city center (such as stations 12, 37, 100), commercial complexes (such as stations 17, 19), and comprehensive transportation hubs (such as stations 35). The passenger flow is mainly for shopping, tourism, and other leisure and entertainment. It should be noted that station 0 is the starting station of line 1, and there are many residential agglomeration areas around this station, with scenic spots and ancient towns nearby.

Therefore, the passenger flow from station 0 to station 35 and from station 35 to station 0 is usually on weekends.

Table 10. Details of the 10 largest groups of OD passenger flow within 1 h on weekends (April 24).

Serial Number	Hourly OD Passenger Flow (Person/h)	Starting Station Number	Arrival Station Number
1	282	100	35
2	280	35	100
3	131	17	19
4	127	0	35
5	127	35	37
6	121	19	17
7	105	17	12
8	104	35	0
9	103	12	17
10	101	37	35

Additionally, this paper selects four pairs of passenger traffic flows from the origin and destination stations, respectively, to perform a comparison between the actual value and the three predicted values of HA, Conv LSTM, and GCN-GRU for a total of 11 days, or 264 passenger flow values. As shown in Figures 12–15, by fitting and comparing the predicted ridership values with the actual observed values, the GCN-GRU model well captures the spatial–temporal nonlinear characteristics of passenger flow and can adapt to real-time ridership changes.

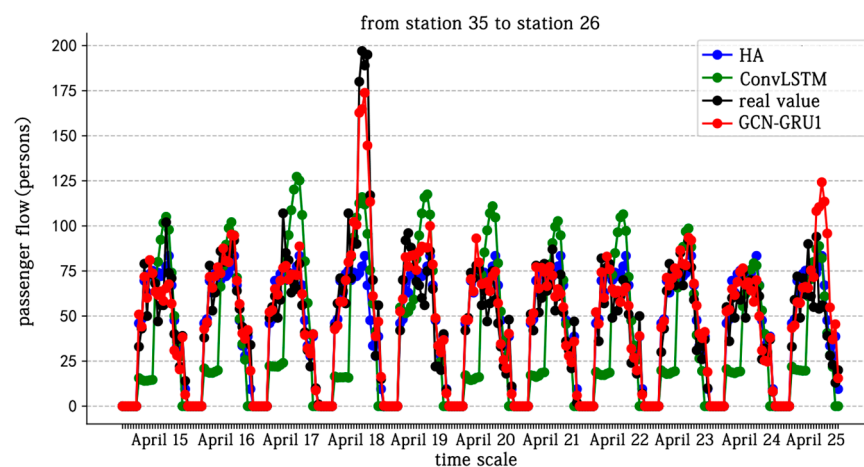


Figure 12. The OD passenger flow prediction effect picture of various models from station 35 to station 26.

Figure 12 shows the change in passenger traffic flow over time from the normal train station to the high-speed train station in the city. The normal train station in the city is located in the old city center, and the high-speed train station is located in the north of the city. There is no obvious morning and evening peak in this pair of OD, and the number of ridership during the day is much higher than that at night.

Figure 13 shows the passenger traffic flow from the starting station of Line 1 to the train station in the city. There are obvious morning and evening peaks in this pair of OD ridership, and the evening peak on Sunday (such as April 18) and the morning peak on Monday (April 19) are both obvious compared with other periods, which is closely related to the city's involvement in the integrated development of the Yangtze River Delta, and passengers work and travel across the city.

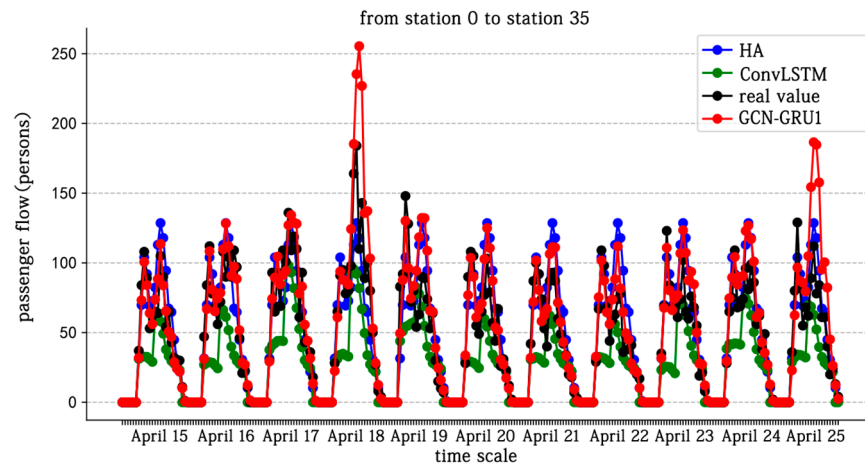


Figure 13. The OD passenger flow prediction effect picture of various models from station 0 to station 35.

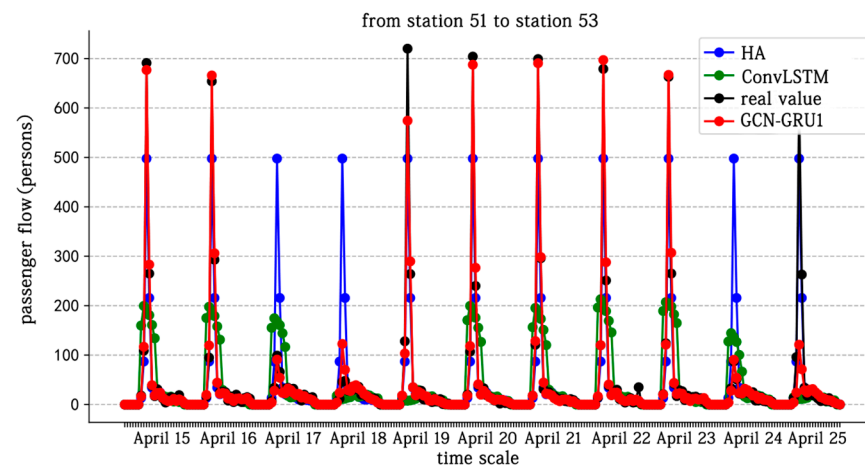


Figure 14. The OD passenger flow prediction effect picture of various models from station 51 to station 53.

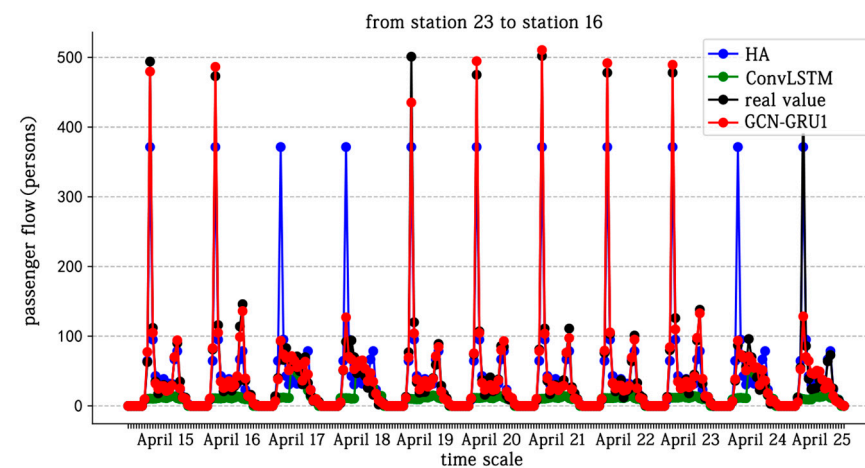


Figure 15. The OD passenger flow prediction effect picture of various models from station 23 to station 16.

Figure 14 shows the passenger traffic flow from the residential agglomeration area to the office agglomeration area in the city, which is obvious in the morning peak and closely related to the commuting ridership. There are fewer commuters from home to the company or other travel ridership in the evening peak, so the evening peak is not obvious.

Figure 15 shows the passenger traffic flow from the residential agglomeration area to the office agglomeration area on the busiest line, line 1, of the city's rail transit. The morning and evening peaks are both obvious, and the passenger flow in the morning peak is much larger than that in the evening peak, which is mainly related to passenger traffic flow in the morning peak. Because station 16 is located in the core position of a certain area of the city, there are many places of leisure and entertainment nearby, and the passenger flow for leisure and entertainment is mainly in the evening peak.

To sum up, the GCN-GRU model shows a superior prediction effect and better applicability. The above experiments also verify that the GCN-GRU model proposed in this paper can well capture the spatial-temporal nonlinear characteristics of large-scale OD passenger flows in urban rail transit, adapt to changes in passenger flow, and improve prediction accuracy while ensuring the simplicity of the model structure. Therefore, this network can be used to achieve large-scale OD passenger flow prediction for urban rail transit.

5. Conclusions

OD passenger flow prediction is more challenging than general passenger flow prediction because it involves a huge matrix and a huge amount of data, and the prediction accuracy of a large-scale OD matrix is affected by data sparsity. To solve this problem, the concepts of station neighborhood selection are first determined. On this basis, a GCN network is embedded to capture spatial features, and an advanced weighted aggregator is improved to sense the sparsity and range of the data. Then, the GRU network is used to capture temporal features, and a large-scale OD passenger flow prediction model based on GCN-GRU is constructed. By using the actual OD passenger flow data in an urban rail transit network, it is demonstrated that the model proposed in this paper achieves significantly better performance than the benchmark models. The better performance can be attributed to the fact that GCN captures the spatial characteristics and GRU considers the temporal characteristics of the OD flow of the urban transit network. This study provides essential and more reliable data for adjusting transportation plans, performing train operation control, and coordinating operation organization. The results can also be used to induce passenger travel paths, predict passenger flow congestion, and develop plans for emergency coordination and joint control, thus contributing to the sustainable development of urban rail transit systems.

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